EE/CS 228 - Winter 2026 Midterm Exam

Name (print):

Student ID:

Remarks:

- Any consultation with others will lead to a failing grade in the course and will be reported for disciplinary action. **If you are caught copying someone else's solution, you will FAIL.**
- TOTAL: 30 points—1 point each question.
- Some questions might have **multiple** correct answers. You will get 1 point if you select all correct answers, 0.5 points if you select a subset of the correct answers, and 0 if your selection contains any incorrect answer.
- Please write down your answer in front of each question.
- The last 4 pages are intentionally left blank, which can be removed and used as scratch papers.
- GOOD LUCK!

Problems

Problem 1. Which ones are not typical practices when designing deep nets?

(a) ✓Training with small amount of data to avoid overfitting

(b) Trying out different hyperparameters

(c) Using data augmentation to improve accuracy

(d) Training with early stopping to prevent overfitting

Problem 2. While training a neural network, you notice that the loss/error stops decreasing after few epochs. The reason for this could be: (1) Already reached global minimizer, (2) Regularization parameter is high, (3) Stuck at local minima. Which of these are plausible explanations?

(a) 1 and 2

(b) 1 and 3

(c) 2 and 3

(d) ✓1, 2, and 3

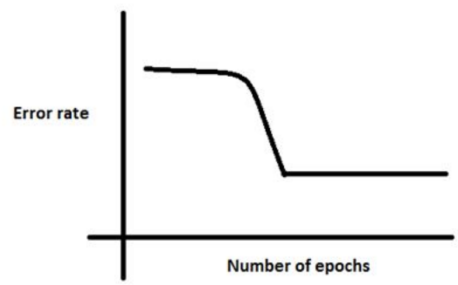

Figure 1: Loss landscape

Problem 3. Which of the followings are true about squared (i.e. least-squares) loss?

(a) ✓Squared loss can achieve exactly zero training loss.

(b) Squared loss does not apply to multiclass classification.

(c) ✓Squared loss is not preferable for classification tasks compared to logistic loss.

(d) All of the above.

Problem 4. Suppose you have a fully-connected network with three hidden layers. Input dimension is 300, and the dimensions of the first, second, and third hidden layers are 200, 200, and 60 respectively. The output layer is 30 dimensional. What is the total number of parameters in this network? (Please assume that **there are also bias variables in each layer**).

(a) !114290 (b) 114060 (c) 113800 (d) 114080

Problem 5. Which of the statements are correct regarding the loss landscape on the right side? Each curve corresponds to a level set where the loss function is equal to a fixed value.

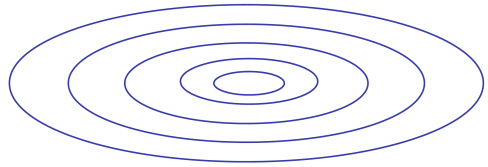

Figure 2: Loss landscape

(a) ✓Gradient along the y-axis is larger than gradient along the x-axis

(b) Optimization along the x-axis is faster than y-axis

(c) SGD is expected to speed up the optimization compared to Adagrad.

(d) ✓Momentum will speed up the optimization compared to SGD

Problem 6. What does the vanishing gradient problem refer to in the context of deep learning?

(a) The issue when the gradient of the loss function becomes too large, causing unstable learning

(b) The situation when gradients are identical for all layers, leading to uniform updates

(c) The scenario where the model stops learning due to early stopping

(d) ✓The problem when gradients backpropagated through deep layers become increasingly small, leading to negligible updates

Problem 7. Suppose $f(w)$ is convex and L-smooth. For gradient descent update $w_{t+1} = w_t - \eta \nabla f(w_t)$ with learning rate $\eta > 0$, one can derive

$$f(w_{t+1}) \leq f(w_t) + \left(\frac{L\eta^2}{2} - \eta \right) \|\nabla f(w_t)\|_2^2.$$

Which of the following choices ensure that the function value is guaranteed to be strictly decreasing?

(a) $!\frac{1}{L}$ (b) $\frac{2}{L}$ (c) $\frac{4}{L}$ (d) $!\frac{1}{2L}$

Problem 8. What are key ideas behind batch normalization in deep learning?

(a) ✓To standardize the inputs to a layer, helping to stabilize learning and reduce problems caused by poor initialization

(b) To normalize the weights of the neural network

(c) ✓To stabilize the training process by normalizing the outputs of each layer

(d) To prevent the model from overfitting by clipping the gradients during backpropagation

Problem 9. What are the benefits of using convolutional layers in a Convolutional Neural Network (CNN) as opposed to fully connected layers?

(a) Convolutional layers must use a fixed 3×3 kernel

(b) ✓Convolutional layers preserve the spatial structure of the input and learn local features, making them suitable for image data

(c) ✓Convolutional layers generally perform better on image data

(d) ✓Convolutional layers typically have less parameters compared to fully connected layers.

Problem 10. What is the advantage of RMSProp optimizer over Adagrad?

(a) RMSProp incorporates momentum term

(b) ✓RMSProp has a more aggressive decay of historical values

(c) RMSProp uses smaller mini-batch sizes

(d) RMSProp uses adaptive moments

Problem 11. Why is Adam optimizer so popular?

(a) Adam uses optimized batch size selection

(b) ✓Adam is adaptive to the loss landscape

(c) No hyperparameters are used in Adam

(d) ✓Adam incorporates momentum term

Problem 12. Consider a problem with $N = 4,096 = 2^{12}$ training examples. We use minibatch SGD with batch size B. You are given that, as a function of B, training requires $200N/\sqrt{B}$ mini-batch iterations to reach satisfactory accuracy. Additionally, if $B \leq 256 = 2^8$, it takes $B^{1/4}$ microseconds to compute gradient for one mini-batch, whereas for $B \geq 256 = 2^8$, it takes $B/64$ microseconds. You choose the optimal batch size to minimize the training time. How much speed up do you achieve over plain SGD with batch size 1?

(a) 2 (b) !4 (c) 8 (d) 16

Problem 13. Consider a problem with $N = 4,096 = 2^{12}$ training examples. We use minibatch SGD with batch size B. You are given that, as a function of B, training requires $200N/\sqrt{B}$ mini-batch iterations to reach satisfactory accuracy. Additionally, if $B \leq 256 = 2^8$, it takes $B^{1/4}$ microseconds to compute gradient for one mini-batch, whereas for $B \geq 256 = 2^8$, it takes $B/64$ microseconds. You choose the optimal batch size to minimize the training time. How much speed up do you achieve over plain gradient descent (batch size $= 4,096$)?

(a) 2 (b) !4 (c) 8 (d) 16

Problem 14. Which of the following statements are correct about data augmentation?

(a) Data augmentation typically speeds up the optimization process.

(b) ✓Aaugmented data point has the same label as the original one.

(c) Data augmentation is a norm-based regularization strategy.

(d) ✓Data augmentation typically improves the accuracy.

Problem 15. Suppose you have a dataset $X = [x_1 \ \dots \ x_n]^T$ and $y = [y_1 \ \dots \ y_n]^T$. Consider the ridge regularized problem $L(\theta) = \|y - X\theta\|_{\ell_2}^2 + \lambda\|\theta\|_{\ell_2}^2$. What is the minimizer of $L(\theta)$?

(a) $X^T(XX^T + \lambda I)^{-1}y$

(b) ✓$(X^TX + \lambda I)^{-1}X^Ty$

(c) $(X^TX + 2\lambda I)^{-1}X^Ty$

(d) $X^T(XX^T + 2\lambda I)^{-1}y$

Problem 16. Which of the followings are true about the max-norm regularization?

(a) ✓Unlike weight-decay which is global, max-norm helps control the weights of the individual neurons.

(b) Unlike L1 penalty which is global, max-norm helps promote the sparsity of the weights of individual neurons.

(c) ✓Max-norm can synergize with other regularization techniques such as dropout.

(d) ✓Suppose the incoming weights (connections) to a neuron are $w = [10, 10, 10, 10]$. If we apply a max-norm constraint of $\|w\|_{\ell_2} \leq 1$, the new weights become $w = [0.5, 0.5, 0.5, 0.5]$.

Problem 17. Let (x, y) be a training example. Consider the loss function $L(\theta) = \frac{1}{2}(y - \sigma(x^T\theta))^2$ where σ is the logistic function $\sigma(x) = \frac{1}{1+e^{-x}}$. What is the gradient of $L(\theta)$?

(a) $(1 - \sigma(x^T\theta))(\sigma(x^T\theta) - y)x$

(b) ✓$\sigma(x^T\theta)(1 - \sigma(x^T\theta))(\sigma(x^T\theta) - y)x$

(c) $\max(0, \sigma(x^T\theta))(\sigma(x^T\theta) - y)x$

(d) $\sigma(\max(0, x^T\theta))(1 - \sigma(\max(0, x^T\theta)))(\sigma(x^T\theta) - y)x$

(e) None of the above

Problem 18. Suppose input tensor to a two layer CNN is $102 \times 102 \times 3$ dimensional. The first convolutional layer consists of 100 filters of size 7×7 and uses stride 5 (no padding). The second layer is a fully connected network with output dimension 10. Recall that this layer flattens the output of the first layer into a vector. What is the total number of parameters in this network? (Assume **no bias variables**)

(a) 18700

(b) 404900

(c) ✓414700

(d) 9039700

(e) None of the above

Problem 19. Consider the following simple neural network

$$
\begin{aligned}
f &= x_1 - 2x_2 \\
x_1 &= 3x_3 - x_4 \\
x_2 &= x_5 - 2x_3 \\
x_3 &= 2x_4 - x_5
\end{aligned}
$$

What is $\partial f / \partial x_4 + \partial f / \partial x_5$?

(a) -3 (b) 2 (c) !4 (d) 7

Problem 20. Consider the following simple neural network

$$
\begin{aligned}
f &= \mathrm{ReLU}(x_1 - 2x_2) \\
x_1 &= \mathrm{ReLU}(3x_3 - x_4) \\
x_2 &= \mathrm{ReLU}(x_5 - 2x_3) \\
x_3 &= \mathrm{ReLU}(2x_4 - x_5)
\end{aligned}
$$

What is $\partial f / \partial x_4 + \partial f / \partial x_5$ evaluated at $x_4 = 1, x_5 = -1$?

(a) 1

(b) ✓2

(c) 3

(d) None of above

Problem 21. Consider the fully connected network $f(x) = v^T \text{ReLU}(W_2 \text{ReLU}(W_1 x))$. Suppose

$$W_1 = \begin{bmatrix} 2 & 1 \\ 1 & -2 \\ 1 & 0 \end{bmatrix} \quad W_2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \end{bmatrix} \quad \text{and} \quad v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

What is the output evaluated at $x = [1, 2]$?

(a) -3

(b) 3

(c) ✓6

(d) 9

Problem 22. Consider a network $f(x) = v^T \text{Conv}(x, k)$. Suppose there is no zero padding. Here x is a 5×1 vector, k is a 3×1 filter (with stride 1) and v is a 3×1 vector. What is the output of the network at $x = [3, 2, 1, 2, 3]^T$ and $v = [2, -1, -2]^T, k = [0, 1, 2]^T$? Note: Don't flip k—we are actually computing cross-correlation rather than convolution.

(a) 3 (b) 0 (c) -3 (d) -9 (e) $!-13$

Problem 23. Consider a network $f(x) = v^T \text{Conv}(x, k)$. Suppose zero padding (on both sides) is applied to keep the output dimension same. Here x is a 5×1 vector, k is a 3×1 filter (with stride 1) and v is a 5×1 vector. What is the output of the network at $x = [3, 2, 1, 2, 3]^T$ with $v = [1, 2, -1, -2, 1]^T, k = [0, 1, 2]^T$? Note: Don't flip k—we are actually computing cross-correlation rather than convolution.

(a) 3 (b) 0 (c) $!-3$ (d) -9 (e) -13

Problem 24. Consider a network $f(x) = v^T \text{max-pool}(x)$. Here, x is a 7×1 vector, the max-pool operation has a kernel of size 3×1 (with stride 1, and no padding), and v is a 5×1 vector. What is the output of the network at $x = [5, 4, 3, 2, 3, 4, 5]^T$ and $v = [1, 1, 2, 1, 1]^T$?

(a) 18 (b) !24 (c) 30 (d) 36

Problem 25. Suppose input tensor to a convolutional layer is $481 \times 361 \times 3$ dimensional. The convolutional layer consists of 30 filters of size 7×7 and uses stride 4 and zero-padding 1. The convolutional layer is followed by a 2×2 average pooling layer with stride 2 (no padding). What is the output dimension? Note: Zero-padding of 1 increases the dimension by two e.g., 4×4 becomes 6×6.

(a) $120 \times 90 \times 30$

(b) $80 \times 60 \times 30$

(c) ✓$60 \times 45 \times 30$

(d) $40 \times 30 \times 30$

Problem 26. Suppose the input tensor to a convolutional layer is $481 \times 361 \times 3$ dimensional. The convolutional layer consists of 30 filters of size 7×7 and uses stride 4 and zero-padding 1. The convolutional layer is followed by a 2×2 average pooling layer with stride 2 (no padding). How many trainable parameters does the convolutional layer have (assuming no bias terms)?

(a) ✓$7 \times 7 \times 3 \times 30$

(b) $7 \times 7 \times 3 \times 30 + 30$

(c) $7 \times 7 \times 30 \times 30 + 30$

(d) $7 \times 7 \times 3 + 30$

Problem 27. Consider a linear network $f(x) = \sum_{i=1}^{n} w_i x_i$, where $w_i \sim \mathcal{N}(0, \sigma^2)$ are independently initialized weights and $x_i \sim \mathcal{N}(0, \gamma^2)$ are independently drawn inputs. Suppose x_i and w_i are independent. What value of $\sigma > 0$ should one choose to make sure $\mathrm{Var}(f(x)) = \gamma^2$?

(a) $\sigma = \frac{1}{n}$ (b) $!\sigma = \frac{1}{\sqrt{n}}$ (c) $\sigma = \frac{2}{n}$ (d) $\sigma = \frac{2}{\sqrt{n}}$

Problem 28. Which of the following statements are true about the ReLU activation?

(a) ReLU outputs values strictly between 0 and 1

(b) ReLU is not differentiable for negative inputs

(c) ✓ReLU helps mitigate the vanishing gradient problem compared to sigmoid function

(d) ReLU doesn't work with SGD

Problem 29. Consider the fully connected network $f_{v,W}(x) = v^T \mathrm{ReLU}(Wx)$. Suppose

$$W = \begin{bmatrix} 2 & -2 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Define the loss function $L(v, W) = \frac{1}{2}(y - f_{v,W}(x))^2$ on the training example $(x = [1,\ -1]^T,\ y = 1)$. What is the partial derivative $\partial L(v, W)/\partial W$?

(a) ✓$\begin{bmatrix} 3 & -3 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & -3 \\ 2 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 6 & -6 \\ 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 6 & -6 \\ 4 & 4 \end{bmatrix}$

(e) None of the above

Problem 30. Consider a training example (x, y) where $y = 3$ i.e., the output label is the third class. For which logits (i.e., output scores of the network) $f(x)$, do we achieve the smallest cross-entropy loss?

(a) $[1, 2, 3]$

(b) $[10, 10, 10]$

(c) ✓$[10, 20, 30]$

(d) $[-100, -200, -300]$

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